

# Lecture 12: KKT First-Order Conditions for Constrained Optimization Problems

MA3236 Non-Linear Programming

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# Quick review for unconstrained optimization

- **Theory:** for  $\mathbf{x}^*$  to be a local minimizer,
  - necessary conditions (filter):  $\nabla f(\mathbf{x}^*) = \mathbf{0}$  and  $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$ ;
  - sufficient conditions (certificate):  $\nabla f(\mathbf{x}^*) = \mathbf{0}$  and  $\mathbf{H}_f(\mathbf{x}^*) \succ 0$ .
- **Algorithms:** to numerically seek stationary points ( $\nabla f(\mathbf{x}^*) = \mathbf{0}$ ), within the line-search iterative framework,
  - directions: steepest descent, Newton, conjugate gradient;
  - step sizes: one-dimensional search—bisection, Newton, golden section.

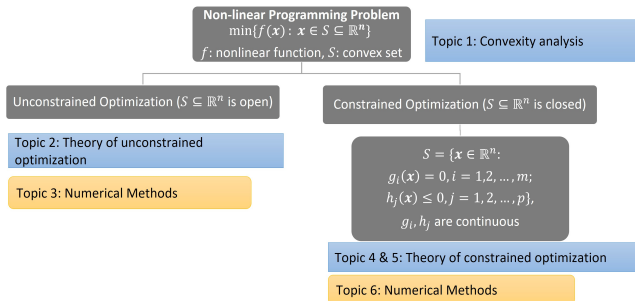
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  - directions: steepest descent, Newton, conjugate gradient;
  - step sizes: one-dimensional search—bisection, Newton, golden section.
- **Backbone:** every perturbation direction in  $\mathbf{R}^n$  is admissible.

For the second half of this course

Directions violating constraints should be removed and  $\nabla f(\mathbf{x}^*)$  need not vanish. [Theory?](#) [Algorithms?](#)

# Topics



In Topic 4, we study optimality conditions for **constrained optimization**.

- KKT first-order necessary conditions and constraint qualifications
- KKT second-order necessary and sufficient conditions
- KKT first-order necessary and sufficient conditions for convex optimization and SVM

**KKT: Karush (1939)–Kuhn–Tucker (1951).**

## Recap: the smooth constrained problem

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}), \quad \mathcal{S} := \left\{ \mathbf{x} \in \mathcal{D} : \begin{array}{ll} g_i(\mathbf{x}) = 0, & i = 1, \dots, m, \\ h_j(\mathbf{x}) \leq 0, & j = 1, \dots, p \end{array} \right\}.$$

- $\mathcal{D} \subseteq \mathbb{R}^n$  is open.
- $f, g_i, h_j : \mathcal{D} \rightarrow \mathbb{R}$  are continuously differentiable.
- $g_i$ 's: equality constraints;  $h_j$ 's: inequality constraints.

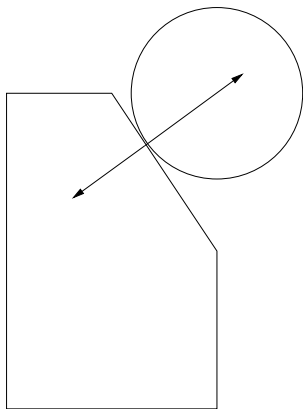
A point  $\mathbf{x}^*$  is a **local minimizer** if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{N}_* \cap \mathcal{S},$$

with  $\mathcal{N}_*$  a neighborhood of  $\mathbf{x}^*$  in  $\mathcal{D}$ ; if  $\forall \mathbf{x} \in \mathcal{S}$ , then a **global minimizer**.

# A boundary optimum from the earlier L02 and L04

$$\begin{array}{ll}\min_{\mathbf{x} \in \mathbb{R}^2} & (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{s.t.} & x_1 \leq 4, \quad x_2 \leq 6, \quad 3x_1 + 2x_2 \leq 18, \quad x_1, x_2 \geq 0.\end{array}$$



The closest feasible point to  $(4, 6)$  is

$$\mathbf{x}^* = \left( \frac{34}{13}, \frac{66}{13} \right)^T.$$

Only  $3x_1 + 2x_2 \leq 18$  is active,

$$\nabla f(\mathbf{x}^*) = \left( -\frac{36}{13}, -\frac{24}{13} \right)^T \neq \mathbf{0}.$$

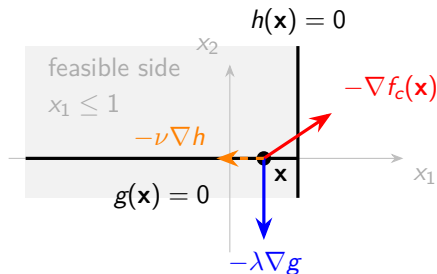
The active inequality prevents further objective reduction.

# One common “physical” system

Let  $\mathbf{x} = (x_1, x_2)$  and consider

$$f_c(\mathbf{x}) = \frac{1}{2}(x_1 - c)^2 - x_2, \quad g(\mathbf{x}) = x_2 = 0, \quad h(\mathbf{x}) = x_1 - 1 \leq 0.$$

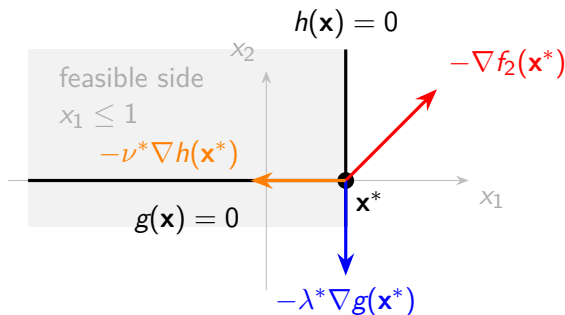
- $f_c$ : potential energy of a horizontal spring anchored at  $x_1 = c$ , together with a constant upward load.
- $g = 0$ : a frictionless horizontal rail.
- $h \leq 0$ : a rigid wall at  $x_1 = 1$ ; the particle must stay to its left.



$$\begin{aligned} \mathbf{F}_f &:= -\nabla f_c, \\ \mathbf{F}_g &:= -\lambda \nabla g, \quad \lambda \in \mathbf{R}, \\ \mathbf{F}_h &:= -\nu \nabla h, \quad \nu \geq 0. \end{aligned}$$

# Active wall: three forces balance

Take  $c = 2$ . On the feasible half-line, the minimizer is  $\mathbf{x}^* = (1, 0)$ .



$$-\nabla f_2(\mathbf{x}^*) = (1, 1)^T, \quad -\nabla g(\mathbf{x}^*) = (0, -1)^T, \quad -\nabla h(\mathbf{x}^*) = (-1, 0)^T.$$

$$\Rightarrow -\nabla f_2(\mathbf{x}^*) - \lambda^* \nabla g(\mathbf{x}^*) - \nu^* \nabla h(\mathbf{x}^*) = \mathbf{0}, \quad \lambda^* = \nu^* = 1.$$



# From force balance to KKT first-order stationarity

The force picture suggests the following balance at a candidate point  $\mathbf{x}^*$ :

$$-\nabla f(\mathbf{x}^*) - \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) - \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0},$$
$$\lambda_i^* \in \mathbb{R}, \quad \nu_j^* \geq 0.$$

Note that “ $\nu_j \geq 0$ ” applies to the canonical form  $h_j(\mathbf{x}) \leq 0$  in minimization.

Equivalently, after multiplying by  $-1$ ,

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

# Lagrange multipliers and the Lagrangian

The coefficients  $\lambda_i$  and  $\nu_j$  are the *Lagrange multipliers*. They measure the strengths of the constraint reactions in the force balance. Define

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) := f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x})$$

as the *Lagrangian function*. Then force balance is simply

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}.$$

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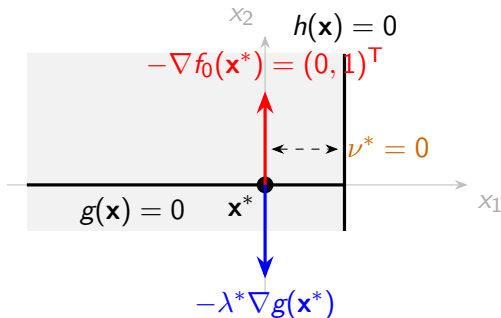
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$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}.$$

Is it done? No! Wall can be inactive.

# Inactive wall: clearance means zero wall force

Move the spring anchor to  $c = 0$ . The minimizer is  $\mathbf{x}^* = (0, 0)$  and  $h(\mathbf{x}^*) = -1 < 0$ .



$$-\nabla f_0(\mathbf{x}^*) - \lambda^* \nabla g(\mathbf{x}^*) - \cancel{\nu^* \nabla h(\mathbf{x}^*)} = \mathbf{0}, \quad \lambda^* = 1, \quad \nu^* = 0.$$

# Active set and complementary slackness

At a feasible point  $\mathbf{x}$ , define the *active set*

$$J(\mathbf{x}) := \{j : h_j(\mathbf{x}) = 0\}.$$

| wall status        | constraint value        | possible force coefficient |
|--------------------|-------------------------|----------------------------|
| active/contact     | $h_j(\mathbf{x}^*) = 0$ | $\nu_j^* \geq 0$           |
| inactive/clearance | $h_j(\mathbf{x}^*) < 0$ | $\nu_j^* = 0$              |

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Both cases are encoded by

$$\nu_j^* \geq 0, \quad h_j(\mathbf{x}^*) \leq 0, \quad \boxed{\nu_j^* h_j(\mathbf{x}^*) = 0}.$$

This is *complementary slackness*: either the wall has slack, or it may carry load.

# KKT first-order conditions

## Definition 12.1

A feasible point  $\mathbf{x}^* \in \mathcal{S}$  is a *KKT point*—that is, a point satisfying the *KKT first-order conditions*—if there exist  $\boldsymbol{\lambda}^* \in \mathbb{R}^m$  and  $\boldsymbol{\nu}^* \in \mathbb{R}^p$  such that

$$\text{stationarity: } \nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0},$$

$$\text{primal feasibility: } g_i(\mathbf{x}^*) = 0, \quad h_j(\mathbf{x}^*) \leq 0,$$

$$\text{dual feasibility: } \nu_j^* \geq 0,$$

$$\text{complementary slackness: } \nu_j^* h_j(\mathbf{x}^*) = 0.$$

By using the active set, stationarity is equivalent to

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}, \quad \nu_j^* \geq 0.$$

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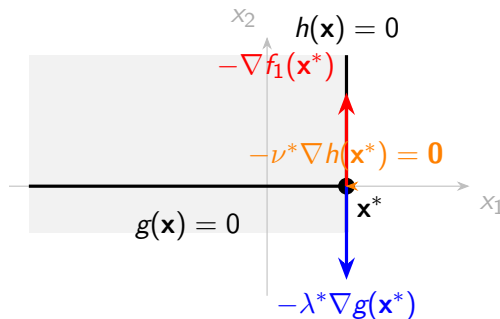
$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}, \quad \nu_j^* \geq 0.$$

$$h_j(\mathbf{x}^*) = 0 \Rightarrow \nu_j^* > 0?$$



# Active but unloaded contact

Take the same physical system with  $c = 1$ . Then  $\mathbf{x}^* = (1, 0)$  touches the wall, but the spring exerts no horizontal force.



$$h(\mathbf{x}^*) = 0, \quad \nu^* = 0, \quad \lambda^* = 1.$$

Complementary slackness allows this *active but unloaded* case.

# The three contact cases in one model

$$f_c(\mathbf{x}) = \frac{1}{2}(x_1 - c)^2 - x_2, \quad g(\mathbf{x}) = x_2, \quad h(\mathbf{x}) = x_1 - 1.$$

| $c$ | minimizer $\mathbf{x}^*$ | wall status         | $\nu^*$ |
|-----|--------------------------|---------------------|---------|
| 2   | (1, 0)                   | active and loaded   | 1       |
| 0   | (0, 0)                   | inactive            | 0       |
| 1   | (1, 0)                   | active but unloaded | 0       |

## One-way implication only

$$\nu_j^* > 0 \implies h_j(\mathbf{x}^*) = 0, \quad h_j(\mathbf{x}^*) = 0 \not\implies \nu_j^* > 0.$$

*Strict complementary slackness*: “active but unloaded” is impossible.

$$h_j(\mathbf{x}^*) = 0, \nu_j^* > 0 \quad \text{or} \quad h_j(\mathbf{x}^*) < 0, \nu_j^* = 0.$$

## Return to the opening example: one active wall

Write all five inequalities as

$$\begin{aligned}h_1 &= x_1 - 4, & h_2 &= x_2 - 6, & h_3 &= 3x_1 + 2x_2 - 18, \\h_4 &= -x_1, & h_5 &= -x_2.\end{aligned}$$

At  $\mathbf{x}^* = (34/13, 66/13)^\top$ ,

$$J(\mathbf{x}^*) = \{3\}, \quad \nabla f(\mathbf{x}^*) = -\frac{12}{13}(3, 2)^\top.$$

Hence active-set stationarity gives

$$\nabla f(\mathbf{x}^*) + \nu_3^*(3, 2)^\top = \mathbf{0} \quad \implies \quad \nu_3^* = \frac{12}{13}.$$

Set  $\nu_j^* = 0$  for  $j \neq 3$ .

# Two questions hidden by the force picture

## Question 1: existence

Does every constrained local minimizer  $\mathbf{x}^*$  satisfy the KKT first-order conditions—that is, admit suitable multipliers  $\boldsymbol{\lambda}^* \in \mathbf{R}^m$  and  $\boldsymbol{\nu}^* \in \mathbf{R}_+^p$ ?

## Question 2: uniqueness

If multipliers  $\boldsymbol{\lambda}^* \in \mathbf{R}^m$  and  $\boldsymbol{\nu}^* \in \mathbf{R}_+^p$  satisfying the KKT first-order conditions exist, are they unique?

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Both answers are **NO** without a **regularity assumption** on the local force environment.

# Pathology 1: a minimizer violating the KKT

$$\min_{(x_1, x_2) \in \mathbb{R}^2} f(\mathbf{x}) = x_1 + x_2^2 \quad \text{s. t.} \quad h(\mathbf{x}) = x_1^2 \leq 0.$$

The feasible set is the line  $x_1 = 0$ , so  $\mathbf{x}^* = (0, 0)$  is its unique global minimizer. However,

$$\nabla f(\mathbf{x}^*) = (1, 0)^T, \quad \nabla h(\mathbf{x}^*) = \mathbf{0}.$$

Stationarity would require

$$(1, 0)^T + \nu \mathbf{0} = \mathbf{0},$$

which is impossible for every  $\nu \geq 0$ .

## Pathology 2: two opposing walls, many multipliers

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = x_1 + x_2^2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = x_1 \leq 0, \quad h_2(\mathbf{x}) = -x_1 \leq 0.$$

The two walls squeeze the particle onto  $x_1 = 0$ ; again  $\mathbf{x}^* = (0, 0)$  is the unique global minimizer. Both constraints are active and

$$(1, 0)^T + \nu_1(1, 0)^T + \nu_2(-1, 0)^T = \mathbf{0}.$$

Hence

$$(\nu_1, \nu_2) = (t, 1 + t), \quad t \geq 0.$$

The active normals are collinear, so the same net force has infinitely many nonnegative decompositions.

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Lesson: mathematically equivalent  $\nRightarrow$  numerically equivalent.



# LICQ: a regular force environment

## Definition 12.2

The *linear independence constraint qualification (LICQ)* holds at a feasible point  $\mathbf{x}$  if

$$\{\nabla g_i(\mathbf{x}) : i = 1, \dots, m\} \cup \{\nabla h_j(\mathbf{x}) : j \in J(\mathbf{x})\}$$

is linearly independent.

- no active surface normal is zero:

$$\text{Pathology 1 : } \nabla h(\mathbf{x}^*) = \mathbf{0};$$

- no active normal is redundant, i.e., active surfaces meet transversally:

$$\text{Pathology 2 : } \nabla h_1(\mathbf{x}^*) \parallel \nabla h_2(\mathbf{x}^*).$$

## LICQ answers Question 2: uniqueness

Suppose two multiplier pairs satisfy the KKT first-order conditions at the same point. Inactive multipliers are zero, and subtraction of their stationarity equations gives

$$\sum_{i=1}^m (\lambda_i^* - \hat{\lambda}_i^*) \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} (\nu_j^* - \hat{\nu}_j^*) \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

Under LICQ, linear independence forces every coefficient to vanish:

$$\lambda_i^* = \hat{\lambda}_i^*, \quad \nu_j^* = \hat{\nu}_j^*.$$

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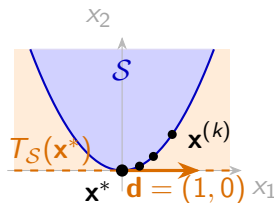
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How about question 1 (existence)?

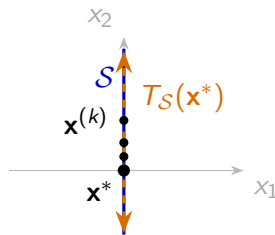
# Actual feasible motion: the tangent cone

For  $\mathbf{x}^* \in \mathcal{S}$ , define the *tangent cone*

$$T_{\mathcal{S}}(\mathbf{x}^*) := \left\{ \mathbf{d} : \begin{array}{l} \exists t^{(k)} \downarrow 0, \mathbf{x}^{(k)} \in \mathcal{S}, \mathbf{x}^{(k)} \rightarrow \mathbf{x}^*, \\ (\mathbf{x}^{(k)} - \mathbf{x}^*)/t^{(k)} \rightarrow \mathbf{d} \end{array} \right\}.$$



$$\mathbf{x}^* = \mathbf{0}, h(\mathbf{x}) = x_1^2 - x_2 \leq 0, \nabla h(\mathbf{x}^*) \neq \mathbf{0}, \\ T_{\mathcal{S}}(\mathbf{x}^*) = \{\mathbf{d} : d_2 \geq 0\}.$$



$$\mathbf{x}^* = \mathbf{0}, h(\mathbf{x}) = x_1^2 \leq 0, \nabla h(\mathbf{x}^*) = \mathbf{0}, \\ T_{\mathcal{S}}(\mathbf{x}^*) = \{\mathbf{d} : d_1 = 0\}.$$

# A local minimizer has no tangent descent direction

## Theorem 12.3

*If  $\mathbf{x}^*$  is a local minimizer and  $f$  is differentiable at  $\mathbf{x}^*$ , then*

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0 \quad \text{for every } \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*).$$

## Proof with Taylor idea.

For a feasible sequence  $\{\mathbf{x}^{(k)}\} \subseteq \mathcal{S}$  realizing  $\mathbf{d}$ ,

$$0 \leq f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) = \nabla f(\mathbf{x}^*)^T (\mathbf{x}^{(k)} - \mathbf{x}^*) + o(\|\mathbf{x}^{(k)} - \mathbf{x}^*\|).$$

Divide by the same  $t^{(k)}$  from the tangent-cone definition. □

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The condition is unconditional and geometric, but  $T_{\mathcal{S}}(\mathbf{x}^*)$  is generally difficult to compute directly.

# A computable alternative: linearized cone

Differentiating the equalities, active inequalities gives the *linearized cone*

$$T_{\text{lin}}(\mathbf{x}) := \left\{ \mathbf{d} : \begin{array}{l} \nabla g_i(\mathbf{x})^\top \mathbf{d} = 0, \quad i = 1, \dots, m, \\ \nabla h_j(\mathbf{x})^\top \mathbf{d} \leq 0, \quad j \in J(\mathbf{x}) \end{array} \right\}.$$

---

For every feasible point,

$$T_S(\mathbf{x}) \subseteq T_{\text{lin}}(\mathbf{x}).$$

Indeed, if a feasible sequence realizes  $\mathbf{d}$ , Taylor expansion gives

$$\frac{g_i(\mathbf{x}^{(k)}) - g_i(\mathbf{x})}{t^{(k)}} \rightarrow \nabla g_i(\mathbf{x})^\top \mathbf{d} = 0,$$

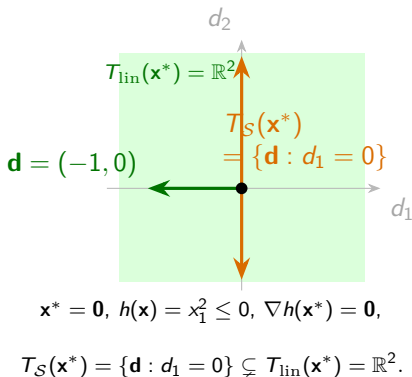
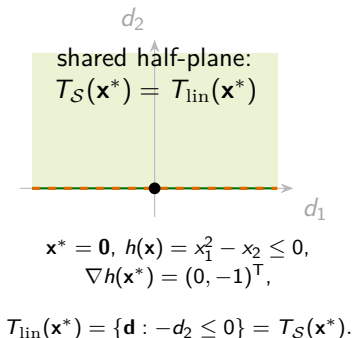
and, for  $j \in J(\mathbf{x})$ ,

$$\frac{h_j(\mathbf{x}^{(k)}) - h_j(\mathbf{x})}{t^{(k)}} \leq 0 \quad \implies \quad \nabla h_j(\mathbf{x})^\top \mathbf{d} \leq 0.$$

# Linearized cones can strictly contain tangent cones

orange: tangent cone

green: linearized cone





# LICQ makes the two cones agree

## Theorem 12.4

*If LICQ holds at  $\mathbf{x} \in S$ , then*

$$T_S(\mathbf{x}) = T_{\text{lin}}(\mathbf{x}).$$

# LICQ makes the two cones agree

## Theorem 12.4

If LICQ holds at  $\mathbf{x} \in \mathcal{S}$ , then

$$T_{\mathcal{S}}(\mathbf{x}) = T_{\text{lin}}(\mathbf{x}).$$

Suppose LICQ holds at  $\mathbf{x}^*$ , how to characterize

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0, \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*)?$$

## Lemma 12.5 (Farkas' alternative lemma)

Let  $B \in \mathbf{R}^{n \times s}$ ,  $C \in \mathbf{R}^{n \times m}$ , and  $\mathbf{u} \in \mathbf{R}^n$ . Exactly one of the following holds:

- $\mathbf{u} = B\mathbf{y} + C\mathbf{v}$  for some  $\mathbf{y} \leq \mathbf{0}$  and  $\mathbf{v} \in \mathbf{R}^m$ ;
- there is  $\mathbf{d} \in \mathbf{R}^n$  such that  $\mathbf{u}^T \mathbf{d} < 0$ ,  $B^T \mathbf{d} \leq \mathbf{0}$ , and  $C^T \mathbf{d} = \mathbf{0}$ .

# LICQ answers Question 1: necessity of KKT

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0, \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*).$$

Farkas: either

- $\mathbf{u} = B\mathbf{y} + C\mathbf{v}$  for some  $\mathbf{y} \leq \mathbf{0}$  and  $\mathbf{v} \in \mathbf{R}^m$ ; or
- there is  $\mathbf{d} \in \mathbf{R}^n$  such that  $\mathbf{u}^T \mathbf{d} < 0$ ,  $B^T \mathbf{d} \leq \mathbf{0}$ , and  $C^T \mathbf{d} = \mathbf{0}$ .

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Recall

$$T_{\text{lin}}(\mathbf{x}^*) = \{\mathbf{d} : G^T \mathbf{d} = \mathbf{0}, H^T \mathbf{d} \leq \mathbf{0}\}$$

with

$$G := [\nabla g_i(\mathbf{x}^*)]_{i=1}^m \in \mathbf{R}^{n \times m}, \quad H := [\nabla h_j(\mathbf{x}^*)]_{j \in J(\mathbf{x}^*)} \in \mathbf{R}^{n \times |J(\mathbf{x}^*)|}.$$

Set  $\mathbf{u} = \nabla f(\mathbf{x}^*)$ ,  $B = H$ , and  $C = G$ .

# LICQ answers Question 1: necessity of KKT

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0, \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*).$$

Farkas: either

- $\mathbf{u} = B\mathbf{y} + C\mathbf{v}$  for some  $\mathbf{y} \leq \mathbf{0}$  and  $\mathbf{v} \in \mathbf{R}^m$ ; or
- there is  $\mathbf{d} \in \mathbf{R}^n$  such that  $\mathbf{u}^\top \mathbf{d} < 0$ ,  $B^\top \mathbf{d} \leq \mathbf{0}$ , and  $C^\top \mathbf{d} = \mathbf{0}$ .

---

Recall

$$T_{\text{lin}}(\mathbf{x}^*) = \{\mathbf{d} : G^\top \mathbf{d} = \mathbf{0}, H^\top \mathbf{d} \leq \mathbf{0}\}$$

with

$$G := [\nabla g_i(\mathbf{x}^*)]_{i=1}^m \in \mathbf{R}^{n \times m}, \quad H := [\nabla h_j(\mathbf{x}^*)]_{j \in J(\mathbf{x}^*)} \in \mathbf{R}^{n \times |J(\mathbf{x}^*)|}.$$

Set  $\mathbf{u} = \nabla f(\mathbf{x}^*)$ ,  $B = H$ , and  $C = G$ . We obtain the **KKT stationarity**

$$\nabla f(\mathbf{x}^*) + \underbrace{B(-\mathbf{y})}_{\boldsymbol{\nu}^*} + \underbrace{C(-\mathbf{v})}_{\boldsymbol{\lambda}^*} = \mathbf{0}$$

# LICQ answers Question 1: KKT FONC

## Theorem 12.6 (KKT first-order necessary conditions (FONC))

*Let  $\mathbf{x}^* \in \mathcal{S}$  be a local minimizer. If LICQ holds at  $\mathbf{x}^*$ , then  $\mathbf{x}^*$  is a KKT point, and its multipliers are unique.*

$$\begin{array}{c} \text{local minimum} \\ \Downarrow \\ \nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) \end{array}$$

Special case: unconstrained optimization.

# LICQ answers Question 1: KKT FONC

## Theorem 12.6 (KKT first-order necessary conditions (FONC))

*Let  $\mathbf{x}^* \in \mathcal{S}$  be a local minimizer. If LICQ holds at  $\mathbf{x}^*$ , then  $\mathbf{x}^*$  is a KKT point, and its multipliers are unique.*

$$\begin{aligned} & \text{local minimum} \\ & \Downarrow \\ & \nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) \\ & \Downarrow \quad \text{LICQ: } T_{\mathcal{S}} = T_{\text{lin}} \\ & \nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) \end{aligned}$$

Special case: unconstrained optimization.

# LICQ answers Question 1: KKT FONC

## Theorem 12.6 (KKT first-order necessary conditions (FONC))

*Let  $\mathbf{x}^* \in \mathcal{S}$  be a local minimizer. If LICQ holds at  $\mathbf{x}^*$ , then  $\mathbf{x}^*$  is a KKT point, and its multipliers are unique.*

$$\begin{array}{c} \text{local minimum} \\ \Downarrow \\ \nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*) \\ \Downarrow \quad \text{LICQ: } T_{\mathcal{S}} = T_{\text{lin}} \\ \nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) \\ \Updownarrow \quad \text{Farkas} \end{array}$$

KKT FONC: force balance and multiplier signs.

Special case: unconstrained optimization.

# An active-set workflow for KKT FONC

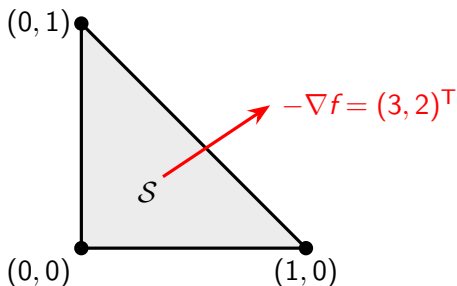
- ① Specify (with  $\mathbf{x}$ ) /guess (without  $\mathbf{x}$ ) the active set  $\hat{J} \subseteq \{1, \dots, p\}$ .
- ② Solve stationarity, all equalities, and  $h_j(\mathbf{x}) = 0$  for  $j \in \hat{J}$ .
- ③ Set  $\nu_j = 0$  for  $j \notin \hat{J}$ .
- ④ Check all omitted inequalities,  $\nu_j \geq 0$ , and complementarity.
- ⑤ Keep the surviving points satisfying the KKT FONC; use additional theory to classify them (next lecture).



# A complete candidate calculation

$$\min_{\mathbf{x} \in \mathbb{R}^2} -3x_1 - 2x_2$$

$$\text{s.t.} \quad h_1 = x_1 + x_2 - 1 \leq 0, \quad h_2 = -x_1 \leq 0, \quad h_3 = -x_2 \leq 0.$$



The linear objective improves in direction  $(3, 2)$ , so the candidate is the vertex  $\mathbf{x}^* = (1, 0)$  with  $J(\mathbf{x}^*) = \{1, 3\}$ .

## A complete candidate calculation (cont'd)

$$\min_{\mathbf{x} \in \mathbb{R}^2} \quad -3x_1 - 2x_2$$

$$\text{s.t.} \quad h_1 = x_1 + x_2 - 1 \leq 0, \quad h_2 = -x_1 \leq 0, \quad h_3 = -x_2 \leq 0.$$

---

At  $\mathbf{x}^* = (1, 0)$ ,

$$\nabla f = (-3, -2)^T, \quad \nabla h_1 = (1, 1)^T, \quad \nabla h_3 = (0, -1)^T.$$

Stationarity gives

$$(-3, -2)^T + \nu_1(1, 1)^T + \nu_3(0, -1)^T = \mathbf{0},$$

hence

$$\nu_1^* = 3, \quad \nu_3^* = 1, \quad \nu_2^* = 0.$$

## Trap 1: KKT FONC are not sufficient

$$\min_{(x_1, x_2)} f(\mathbf{x}) = -x_2 \quad \text{s. t.} \quad h(\mathbf{x}) = x_2 - x_1^2 \leq 0.$$

At  $\mathbf{x}^* = (0, 0)$ ,

$$\nabla f = (0, -1)^T, \quad \nabla h = (0, 1)^T, \quad \nu^* = 1.$$

Thus the KKT FONC hold, and LICQ holds because  $\nabla h(\mathbf{x}^*) \neq \mathbf{0}$ . But the feasible curve  $(t, t^2)$  gives

$$f(t, t^2) = -t^2 < f(0, 0) \quad (t \neq 0).$$

First-order balance cannot distinguish a stable minimum from an unstable equilibrium.

## Trap 2: KKT FONC can hold without LICQ

$$\min_{\mathbf{x} \in \mathbb{R}^2} -x_1 + x_2^2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = x_1 \leq 0, \quad h_2(\mathbf{x}) = 2x_1 \leq 0.$$

The origin is the unique global minimizer. Both constraints are active, but

$$\nabla h_2(\mathbf{0}) = 2\nabla h_1(\mathbf{0}),$$

so LICQ fails. Nevertheless the KKT FONC hold whenever

$$-1 + \nu_1 + 2\nu_2 = 0, \quad \nu_1, \nu_2 \geq 0.$$

LICQ guarantees existence and **uniqueness** of multipliers satisfying the KKT FONC at a local minimizer. Its failure does not imply that those conditions must fail.

# Take-home messages

- Constraints:  $\nabla f = \mathbf{0} \Rightarrow$  balance among the objective force, equality reactions, and active wall reactions.
- Inequalities: one-sided force ( $\nu \geq \mathbf{0}$ ); complementary slackness: contact logic.
- LICQ: algebraic well-posedness of constraint systems (local force environment)  $\Rightarrow$  existence and uniqueness of multipliers.
- Local minimizer + LICQ  $\Rightarrow$  KKT FONC; “ $\Leftarrow$ ”: false.

## Appendix: tangent-cone first-order condition

Choose  $t^{(k)} \downarrow 0$  and feasible  $\mathbf{x}^{(k)}$  such that

$$\frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}} \rightarrow \mathbf{d}.$$

Local optimality and differentiability give

$$0 \leq \frac{f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*)}{t^{(k)}} = \nabla f(\mathbf{x}^*)^\top \frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}} + \frac{o(\|\mathbf{x}^{(k)} - \mathbf{x}^*\|)}{t^{(k)}}.$$

The last term tends to zero because  $\|\mathbf{x}^{(k)} - \mathbf{x}^*\|/t^{(k)}$  is bounded. Taking limits yields

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0.$$

## Appendix: why $T_{\mathcal{S}} \subseteq T_{\text{lin}}$

For every equality constraint,

$$0 = \frac{g_i(\mathbf{x}^{(k)}) - g_i(\mathbf{x})}{t^{(k)}} \rightarrow \nabla g_i(\mathbf{x})^T \mathbf{d}.$$

For an active inequality  $j \in J(\mathbf{x})$ , feasibility gives

$$\frac{h_j(\mathbf{x}^{(k)}) - h_j(\mathbf{x})}{t^{(k)}} \leq 0 \rightarrow \nabla h_j(\mathbf{x})^T \mathbf{d} \leq 0.$$

Therefore every actual feasible velocity satisfies the linearized equalities and inequalities:

$$T_{\mathcal{S}}(\mathbf{x}) \subseteq T_{\text{lin}}(\mathbf{x}).$$

## Appendix: LICQ and the reverse inclusion

Let  $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$  and define

$$I_0 := \{j \in J(\mathbf{x}) : \nabla h_j(\mathbf{x})^\top \mathbf{d} = 0\}.$$

LICQ and the implicit function theorem yield a  $C^1$  curve  $\mathbf{x}(t)$  satisfying

$$g_i(\mathbf{x}(t)) = 0, \quad h_j(\mathbf{x}(t)) = 0 \quad (j \in I_0), \quad \mathbf{x}(0) = \mathbf{x}, \quad \mathbf{x}'(0) = \mathbf{d}.$$

For  $j \in J(\mathbf{x}) \setminus I_0$ ,

$$h_j(\mathbf{x}(t)) = t \nabla h_j(\mathbf{x})^\top \mathbf{d} + o(t) < 0$$

for all sufficiently small  $t > 0$ ; inactive constraints remain strict by continuity. Thus  $\mathbf{x}(t) \in \mathcal{S}$  and  $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x})$ .



## Appendix: the Farkas step

For

$$K := \{\mathbf{d} : G^T \mathbf{d} = \mathbf{0}, H^T \mathbf{d} \leq \mathbf{0}\},$$

the polar cone is

$$K^\circ = \{G\boldsymbol{\lambda} + H\boldsymbol{\nu} : \boldsymbol{\lambda} \in \mathbb{R}^m, \boldsymbol{\nu} \geq \mathbf{0}\}.$$

The no-descent condition says

$$(-\nabla f(\mathbf{x}^*))^T \mathbf{d} \leq 0 \quad \forall \mathbf{d} \in K,$$

so  $-\nabla f(\mathbf{x}^*) \in K^\circ$ . Hence

$$-\nabla f(\mathbf{x}^*) = G\boldsymbol{\lambda} + H\boldsymbol{\nu}, \quad \boldsymbol{\nu} \geq \mathbf{0}.$$

Extend  $\boldsymbol{\nu}$  by zeros on inactive indices to obtain stationarity, dual feasibility, and complementary slackness.